Generate Impulse Responses for a VAR model

This example shows how to generate impulse responses of an interest rate shock on real GDP using `vgxproc`.

Open Script

Load the `Data_USEconModel` data set. This example uses two time series: the logarithm of real GDP, and the real 3-month T-bill rate, both differenced to be approximately stationary. Suppose that a VAR(4) model is appropriate to describe the time series.

```
load Data_USEconModel
DEF = log(DataTable.CPIAUCSL);
GDP = log(DataTable.GDP);
rGDP = diff(GDP - DEF); % Real GDP is GDP - deflation
TB3 = 0.01*DataTable.TB3MS;
dDEF = 4*diff(DEF); % Scaling
rTB3 = TB3(2:end) - dDEF; % Real interest is deflated
Y = [rGDP,rTB3];
```

Define the forecast horizon.

```
FDates = datenum({'30-Jun-2009'; '30-Sep-2009'; '31-Dec-2009';
'31-Mar-2010'; '30-Jun-2010'; '30-Sep-2010'; '31-Dec-2010';
'31-Mar-2011'; '30-Jun-2011'; '30-Sep-2011'; '31-Dec-2011';
'31-Mar-2012'; '30-Jun-2012'; '30-Sep-2012'; '31-Dec-2012';
'31-Mar-2013'; '30-Jun-2013'; '30-Sep-2013'; '31-Dec-2013';
'31-Mar-2014'; '30-Jun-2014' });
FT = numel(FDates);
```

Fit a VAR(4) model specification:

```
Spec = vgxset('n',2,'nAR',4,'Constant',true);
impSpec = vgxvarx(Spec,Y(5:end,:),[],Y(1:4,:));
impSpec = vgxset(impSpec,'Series',...
  {'Transformed real GDP','Transformed real 3-mo T-bill rate'});
```

Generate the innovations processes both with and without an impulse (shock):

```
W0 = zeros(FT, 2); % Innovations without a shock
W1 = W0;
W1(1,2) = sqrt(impSpec.Q(2,2)); % Innovations with a shock
```

Generate the processes with and without the shock:

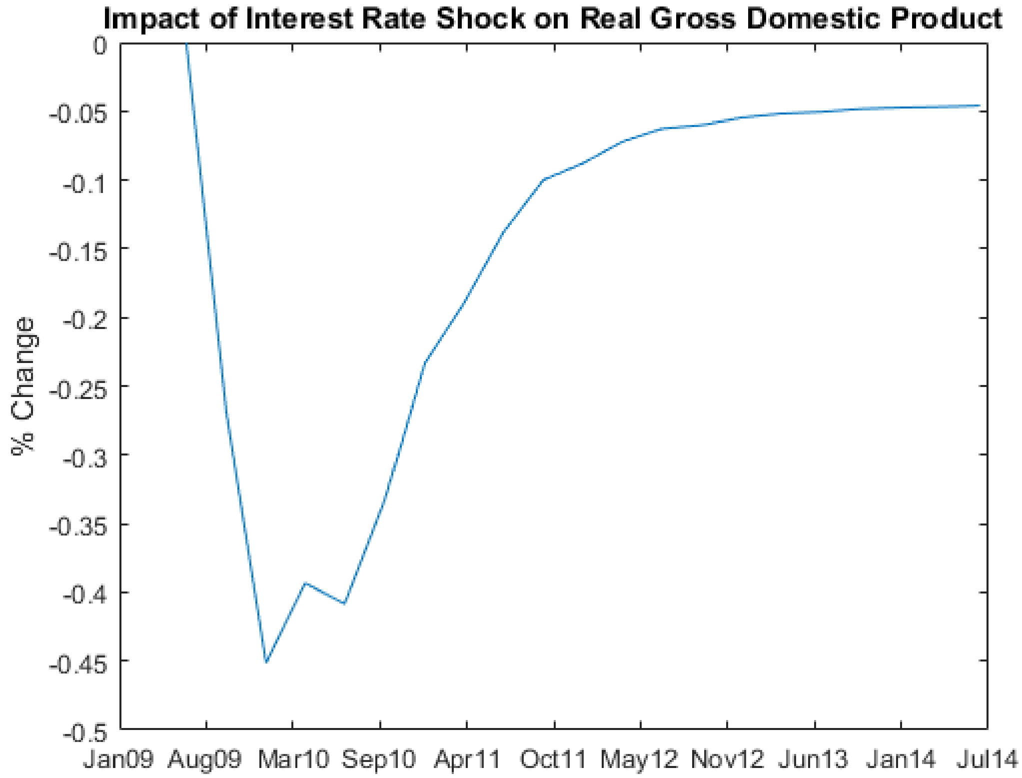
```
Yimpulse = vgxproc(impSpec,W1,[],Y); % Process with shock
Ynoimp = vgxproc(impSpec,W0,[],Y); % Process with no shock
```

Undo the scaling for the GDP processes:

```
Yimp1 = exp(cumsum(Yimpulse(:,1))); % Undo scaling
Ynoimp1 = exp(cumsum(Ynoimp(:,1)));
```

Compute and plot the relative difference between the calculated GDPs:

```
RelDiff = (Yimp1 - Ynoimp1) ./ Yimp1;
plot(FDates,100*RelDiff);dateaxis('x',12)
title(...
'Impact of Interest Rate Shock on Real Gross Domestic Product')
ylabel('% Change')
```



The graph shows that an increased interest rate causes a dip in the real GDP for a short time. Afterwards the real GDP begins to climb again, reaching its former level in about 1 year.

See Also

armairf | vgxpred | vgxproc | vgxsim | vgxvarx

Related Examples

- Fit a VAR Model
- Forecast a VAR Model
- VAR Model Case Study

More About

- Vector Autoregressive (VAR) Models
- VAR Model Forecasting, Simulation, and Analysis